

“AI-Powered Humanoid Robot Fleet Management System”

Team: Robo-Nexus

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GitHub Repository:

<https://github.com/sfsu-joseo/csc675-775-database-api-project-robo-nexus>

Version	Description
V1	Initial Submission

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Normalization Table

Entity	Failed NF	Reason	Fixes
Robot	3NF	Manufacturer depends on robot model id not on robot id, and other ids were creating circular dependency.	remove manufacturer_id, Performance_metric_id, Charging_station_id,Alert_id from robot table.
DeploymentLocation	3NF	Technician_id implies a single one for any deployment location, charging session_id creates circular dependency because chargingSession already references DeploymentLocation.	Remove technician_id ,Charging_Session_id
Technician	3NF	No deployment_location_id means that it fails functional requirements	Add DeploymentLocation_id
Technician	1NF	Name is not an atomic attribute.	Create a different table with names.
SupportRequest	3NF	User_ID is not dependent on any other attribute and is not prime	Removed User_ID

User_Deployme nt_Location	1NF	Missing Primary Key	Added UserDeploymen tLocationID
Emotional_Ana lyze	1NF	Missing Primary Key	Added EmotionalAnal yzeID
PerformanceDi agnostic	1NF	Missing Primary Key	Added PerformanceDi agnosticID
EmotionalInte ractionSessio n	1NF	Missing Primary Key	Added EmotionalInte ractionSessio nID
Technician_Su pport_Request	1NF	Missing Primary Key	Added TechnicianSup portRequestID
RobotCharging Station	1NF	Missing Primary Key	Added RobotCharging StationID
logSession	1NF	Missing Primary Key	Added logSessionID
HardwareMaint enance	1NF	Missing Primary Key	Added HardwareMaint enanceID
ChargingStati onSession	1NF	Missing Primary Key	Added ChargingStati onSessionID
PatientEmotio nalRecord	1NF	Missing Primary Key	Added PatientEmotio nalRecordID

List of Main Data Items and Entities (Normalize)

1. **Robot:** Strong

- 1.1 robot_id: key, numeric (PK)
- 1.2 ~~charging_station_id (FK)~~ (removed)
- 1.3 robotmodel_id: numeric (FK)
- 1.4 deployment_location_id: numeric (FK)
- 1.5 ~~alert_id: numeric (FK)~~ (removed)
- 1.6 user_id: numeric (FK, optional)
- 1.7 software_version_id: numeric (FK)
- 1.8 ~~manufacturer_id: numeric (FK)~~ (removed)
- 1.9 ~~performance_metric_id: numeric (FK)~~ (removed)
- 1.10 robotrole_id: numeric(FK)

2. **RobotModel:** Strong

- 2.1 robotmodel_id: key, numeric (PK)
- 2.2 manufacturer_id: numeric (FK)
- 2.3 model: enum, alphanumeric

3. **User:** Strong

- 3.1. user_id: key, numeric (PK)
- 3.2. user_name: alphanumeric

4. **Manufacturer:** Strong
 - 4.1 manufacturer_id: key, numeric (PK)
 - 4.2 name: enum, alphanumeric
5. **Hardware:** Strong
 - 5.1 hardware_id: key, numeric (PK)
 - 5.2 manufacturer_id: numeric (FK)
6. **RobotHardware:** Weak
 - 6.1 robot_hardware_id: key, numeric
 - 6.2 robot_id: numeric (FK)
7. **SoftwareVersion:** Strong
 - 7.1 software_version_id: key, numeric
8. **DeploymentLocation:** Strong (Superclass)
 - 8.1 deployment_location_id: key, numeric (PK)
 - 8.2 ~~technician_id: numeric (FK)~~
 - 8.3 address_id: numeric (FK)
 - 8.4 environment_type: enum, alphanumeric
(warehouse/hospital/home/factory)
9. **Address:** Strong
 - 9.1 address_id: key, numeric (PK)
 - 9.2 street: alphanumeric
 - 9.3 city: alphanumeric
 - 9.4 state: alphanumeric
 - 9.5 zipcode: alphanumeric
10. **Task:** Strong
 - 10.1 task_id: key, numeric (PK)
 - ~~10.1 ai_response_id: numeric (FK)~~
 - 10.2 task_name: alphanumeric
 - 10.3 description: text
 - 10.4 priority: enum, numeric
 - 10.5 status: bool (Complete)

- 11. **TaskAssignment:** Weak
 - 11.1 task_assignment_id: key, numeric (PK)
 - 11.2 robot_id: numeric (FK)
 - 11.3 user_id: numeric (FK)
 - 11.4 task_id: numeric (FK)
 - 11.5 assigned_at: timestamp
- 12. **TaskExecution:** Weak
 - 12.1 task_execution_id: key, numeric (PK)
 - 12.3 task_id: numeric (FK)
 - 12.4 start_time: timestamp
 - 12.5 end_time: timestamp
 - 12.6 status: bool, (Successful)
- 13. **EmotionalRecord:** Weak
 - 13.1 emotional_id: key, numeric (PK)
 - 13.2 pattern_id: numeric (FK)
 - 13.3 emotional_state: enum, alphanumeric
 - 13.4 timestamp: timestamp
 - 13.5 description: text
- 14. **InteractionSession:** Weak
 - 14.1 interaction_id: key, numeric (PK)
 - 14.2 robot_id: numeric (FK)
 - 14.3 user_id: numeric (FK)
 - 14.4 home_id: numeric (FK)
 - 14.4 start_time: timestamp
 - 14.5 end_time: timestamp
- 15. **RobotRole:** Strong
 - 15.1 robotrole_id: key, numeric (PK)
 - 15.2 user_id: numeric(FK)
 - 15.3 robotrole: enum, alphanumeric
- 16. **EnergyStatus:** Weak
 - 16.1 energy_status_id: key, numeric (PK)

- 16.2 robot_id: numeric (FK)
- 16.3 charging_session_id: numeric (FK)
- 16.4 battery_level: numeric
- 16.5 timestamp: timestamp
- 17. **PerformanceMetric:** Weak
 - 17.1 performance_metric_id: key, numeric (PK)
 - 17.2 robot_id: numeric (FK)
 - 17.3 metric_type: alphanumeric
 - 17.4 metric_value: numeric
 - 17.5 timestamp: timestamp
- 18. **Alert:** Weak
 - 18.1 alert_id: key, numeric (PK)
 - 18.2 robot_id: numeric (FK)
 - 18.3 severity: enum, alphanumeric
 - 18.4 timestamp: timestamp
- 19. **DiagnosticReport:** Weak
 - 19.1 diagnostic_report_id: key, numeric (PK)
 - 19.2 robot_id: numeric (FK)
 - 19.3 ai_model_id: numeric (FK)
 - 19.4 generated_at: timestamp
 - 19.5 summary: text
- 20. **Log:** Weak
 - 20.1 log_id: key, numeric (PK)
 - 20.2 robot_id: numeric (FK)
 - 20.3 ~~deployment_location_id: numeric (FK, optional)~~
 - 20.4 CreatedAt: timestamp
 - 20.5 event_type: alphanumeric
 - 20.6 event_details: text
- 21. **Sensor:** Weak
 - 21.1 sensor_id: key, numeric (PK)
 - 21.2 robot_id: numeric (FK)

- 21.3 sensor_type: alphanumeric
- 21.4 status: enum, alphanumeric
- 22. **ChargingStation**: Strong
 - 22.1 charging_station_id: key, numeric (PK)
 - 22.2 station_name: alphanumeric
 - 22.3 capacity: numeric
 - 22.4 status: enum, alphanumeric
- 23. **ChargingSession**: Weak
 - 23.1 charging_session_id: key, numeric (PK)
 - 23.2 deployment_location_id: numeric (FK)
 - 23.3 start_time: timestamp
 - 23.4 end_time: timestamp
- 24. **SupportRequest**: Weak
 - 24.1 support_request_id: key, numeric (PK)
 - 24.2 robot_id: numeric (FK)
 - 24.3 issue_details: text
 - 24.4 time_reported: timestamp
- 25. **AIModel**: Strong
 - 25.1 aimodel_id: key, numeric (PK)
 - 25.2 model_name: alphanumeric
 - 25.3 service_type: alphanumeric
 - 25.4 version_label: alphanumeric
- 26. **AIRequest**: Weak
 - 26.1 ai_request_id: key, numeric (PK)
 - 26.2 aimodel_id: numeric (FK)
 - 26.3 robot_id: numeric (FK)
 - 26.4 requested_at: timestamp
- 27. **AIResponse**: Weak
 - 27.1 ai_response_id: key, numeric (PK)
 - 27.2 ai_request_id: numeric (FK)

- 27.4 task_id: numeric (FK)
- 27.5 role_id: numeric(FK)

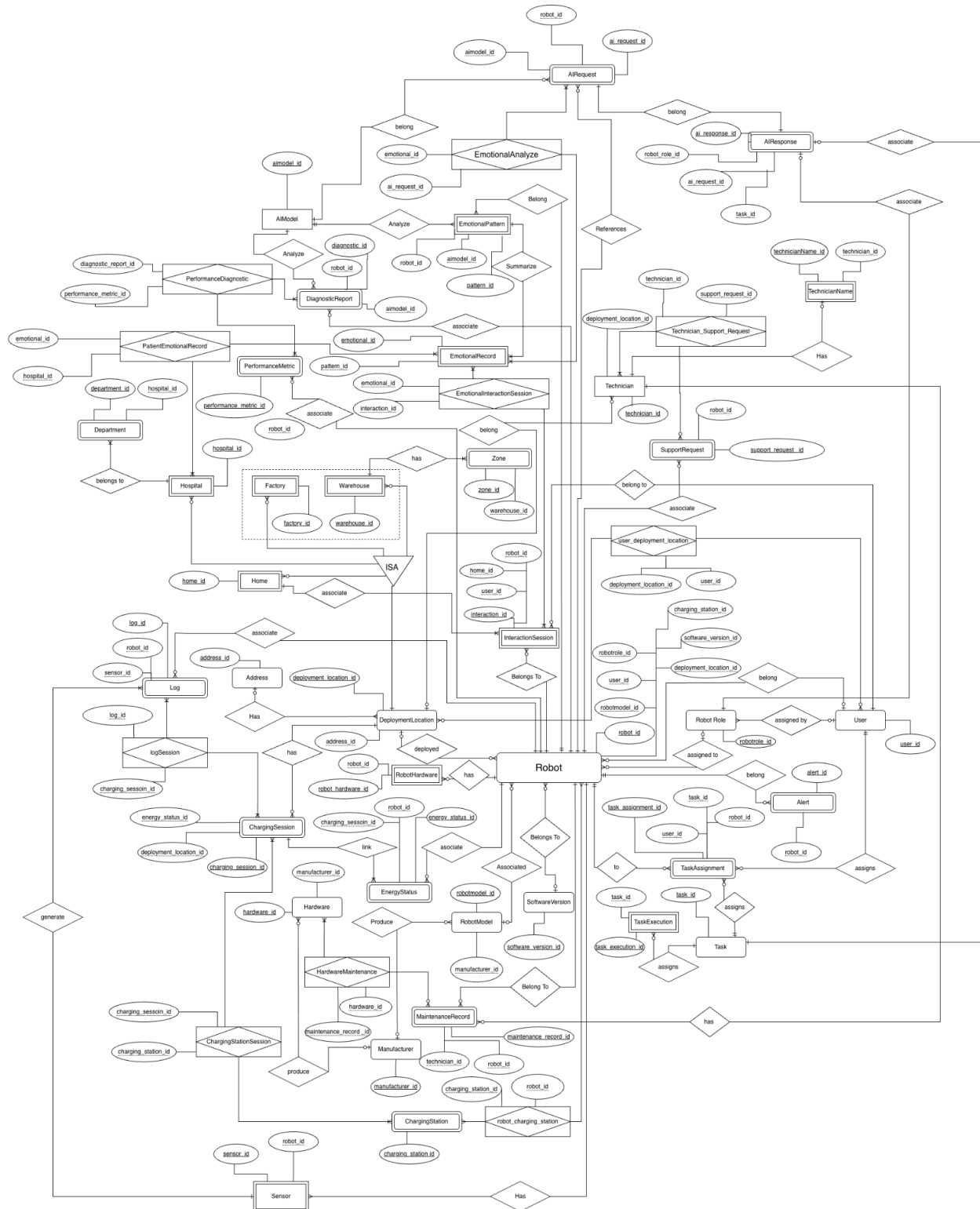
- 27.6 response_payload: text
- 27.7 generated_at: timestamp

- 28. **Hospital:** Weak (Subclass)
 - 28.1 hospital_id: key, numeric (PK)
 - 28.2 deployment_location_id: numeric (FK, unique)
- 29. **Warehouse:** Weak (Subclass)
 - 29.1 warehouse_id: key, numeric (PK)
 - 29.2 deployment_location_id: numeric (FK, unique)
- 30. **Home:** Weak (Subclass)
 - 30.1 home_id: key, numeric (PK)
 - 30.2 deployment_location_id: numeric (FK, unique)
- 31. **Factory:** Weak (Subclass)
 - 31.1 factory_id: key, numeric (PK)
 - 31.2 deployment_location_id: numeric (FK, unique)
- 32. **Department:** Weak
 - 32.1 department_id: key, numeric (PK)
 - 32.2 hospital_id: numeric (FK)
 - 32.3 department_name: alphanumeric
- 33. **EmotionalPattern:** Weak
 - 33.1 emotional_pattern_id: key, numeric (PK)
 - 33.2 robot_id: numeric (FK)
 - 33.3 ai_model_id: numeric (FK)
 - 33.4 pattern_summary: text
- 34. **MaintenanceRecord:** Weak
 - 34.1 maintenance_record_id: key, numeric (PK)
 - 34.2 robot_id: numeric (FK)
 - 34.3 technician_id: numeric (FK)

- 34.4 RecordedAt: timestamp
- 34.5 Description: text
- 35. **Technician**: Strong
 - 35.1 technician_id: key, numeric (PK)
 - 35.2 deployment_location_id (FK)
- 36. TechnicianName: Weak
 - 36.1. TechnicianNameID(PK):key,numeric
 - 36.2. FirstName :Text
 - 36.3. LastName:Text
 - 36.4. TechnicianID(FK)
- 37. **User_Deployment_Location**: Associative
 - 36.1 UserDeploymentLocationID: key, numeric (PK)
 - 36.1 deployment_location_id: numeric(FK)
 - 36.2 user_id: numeric(FK)
- 38. **EmotionalAnalyze**: Associative
 - 37.1 EmotionalAnalyzeID: key, numeric (PK)
 - 37.2 emotional_id: numeric(FK)
 - 37.3 ai_request_id: numeric(FK)
- 39. **PerformanceDiagnostic**: Associative
 - 39.1 PerformanceDiagnosticID: key, numeric (PK)
 - 39.2 diagnostic_report_id: numeric(FK)
 - 39.3 performance_metric_id: numeric(FK)
- 40. **EmotionalInteractionSession**: Associative
 - 40.1 EmotionalInteractionSessionID: key, numeric (PK)
 - 40.2 emotional_id: numeric(FK)
 - 40.3 interaction_id: numeric(FK)
- 41. **Technician_Support_Request**: Associative
 - 41.1 TechnicianSupportRequestID: key, numeric (PK)
 - 41.2 support_request_id: numeric(FK)
 - 41.3 technician_id: numeric(FK)

- 42. **RobotChargingStation**: Associative
 - 42.1 RobotChargingStationID: key, numeric (PK)
 - 42.2 chargin_station_id: numeric(FK)
 - 42.3 robot_id: numeric(FK)
 - 43. **logSession**: Associative
 - 43.1 logSessionID: key, numeric (PK)
 - 43.2 charging_session_id: numeric(FK)
 - 43.3 log_id: numeric(FK)
 - 44. **HardwareMaintenance**: Associative
 - 44.1 HardwareMaintenancelD: key, numeric (PK)
 - 44.2 hardware_id (FK)
 - 44.3 maintainence_record_id (FK)
 - 45. **ChargingStationSession**: Associative
 - 45.1 ChargingStationSessionID: key, numeric (PK)
 - 45.2 charging_session_id: numeric(FK)
 - 45.3 charging_station_id: numeric(FK)
 - 46. **PatientEmotionalRecord**: Associative
 - 46.1 PatientEmotionalRecordID: key, numeric (PK)
 - 46.2 emotional_id: numeric(FK)
 - 46.3 hospital_id: numeric(FK)
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EER Diagram(Normalize)



Database Schema Diagram

